

StudySpace Platform Documentation

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1. Executive Summary

Platform Overview

StudySpace is a community-driven academic resource sharing platform designed to transform how college students access, share, and collaborate on study materials. The platform bridges the gap between scattered academic resources and students who need them, creating a centralized ecosystem for educational content.

Mission

To democratize access to quality study materials by enabling peer-to-peer resource sharing within verified college communities, fostering collaboration, and reducing academic isolation.

Target Audience

- **Primary:** Undergraduate and postgraduate students
- **Secondary:** Faculty members, teaching assistants, and academic institutions
- **Tertiary:** Educational content creators and publishers

Value Proposition

- **For Students:** Access curated, peer-reviewed study materials including notes, PDFs, videos, and previous year questions
- **For Institutions:** A monitored platform that encourages academic collaboration while maintaining content quality
- **For Content Creators:** A distribution channel with community feedback and engagement metrics

Current Scale

- Active deployment for Krishna Institute of Engineering and Technology (KIET)
 - Multi-college architecture ready for horizontal expansion
 - Real-time user tracking and engagement analytics
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2. Problem Statement

The Academic Resource Challenge

Students face significant challenges in accessing quality study materials:

1. Fragmentation of Resources

- Study materials scattered across WhatsApp groups, email chains, and personal drives
- No centralized repository for institutional knowledge
- Previous year questions and notes lost when students graduate

2. Quality Uncertainty

- No mechanism to validate accuracy of shared materials
- Difficulty distinguishing high-quality resources from subpar content
- Lack of peer review systems for academic content

3. Community Isolation

- Students study in silos without collaborative infrastructure
- Limited inter-batch knowledge transfer
- No platform for academic discussions and doubt resolution

4. Information Overload

- Overwhelming amount of unorganized content
- Time wasted searching for relevant materials

- No personalization or filtering mechanisms

Market Gap

Existing solutions focus on either:

- General file sharing (Google Drive, Dropbox) without academic context
- Formal LMS systems (Moodle) without community engagement
- Social networks without educational focus

StudySpace fills this gap by combining:

- Academic-focused file sharing
- Community engagement features
- Quality assurance through voting and moderation

3. Product Features

Core Features

Feature	Description	Status
Resource Library	Upload, browse, and download academic resources (notes, PDFs, videos, PYQs)	Live
Smart Filtering	Filter by semester, branch, subject, chapter, and topic	Live
Community Voting	Upvote/downvote resources to surface quality content	Live
Bookmarking	Save resources for quick access	Live
Following System	Follow content creators and departments	Live

Communication Features

Feature	Description	Status
Chat Rooms	Topic-based discussion rooms (public and private)	Live
Threaded Comments	Reddit-style nested comments on posts and notices	Live
College Notices	Official announcement system for institutions	Live
Notifications	Real-time alerts for follows, replies, and announcements	Live

User Management

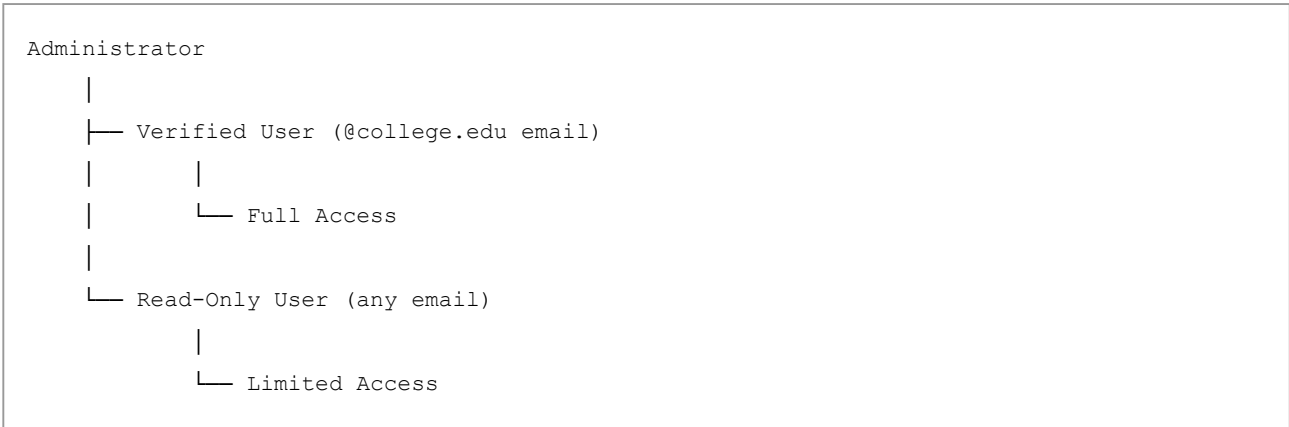
Feature	Description	Status
Profile System	Customizable profiles with bio, photo cropper, and contributions	Live
Role-Based Access	Differentiated permissions for verified and read-only users	Live
User Discovery	Explore and connect with students from same college	Live
Contribution Tracking	Track uploads, votes received, and engagement	Live

Administrative Features

Feature	Description	Status
Room Management	Create, delete, regenerate codes for chat rooms	Live
Member Moderation	Ban/unban members, manage room settings	Live
Notification Controls	Mute notifications per room	Live
Content Moderation	Report and review flagged content	Planned

4. User Roles & Permissions

Role Hierarchy



Permission Matrix

Action	Read-Only User	Verified User	Administrator
View resources	✓	✓	✓
Download resources	✓	✓	✓
Upload resources	✗	✓	✓
Comment on posts	✗	✓	✓
Vote on content	✗	✓	✓
Create chat rooms	✗	✓	✓
Join chat rooms	✗	✓	✓
Delete any content	✗	Own only	✓
Ban users	✗	✗	✓
Access admin panel	✗	✗	✓

Verification Logic

- Verified Users:** Email domain matches registered college (e.g., @kiet.edu)
- Read-Only Users:** Any valid email after college selection
- Administrators:** Manually assigned via database flag

5. System Architecture

Technology Stack

Layer	Technology	Purpose
Frontend	React 18 + Vite	Single Page Application
Styling	Tailwind CSS + shadcn/ui	Component library and styling
Backend	Node.js + Express	API server and business logic
Database	Supabase (PostgreSQL)	Data persistence with RLS
Authentication	Firebase Auth	Identity management
File Storage	Cloudinary	Media and document hosting
CDN/DNS	Cloudflare	Protection and performance
Frontend Hosting	Vercel	Edge deployment
Backend Hosting	Render	Container deployment

Layer Descriptions

Frontend Layer

- **React 18:** Modern UI framework with concurrent features
- **Vite:** Lightning-fast build tool and dev server
- **React Query:** Server state management with caching
- **React Router:** Client-side routing

Backend API Layer

- **Express.js:** RESTful API framework
- **CORS:** Configured for allowed origins only
- **Rate Limiting:** Request throttling per IP
- **Input Validation:** Request sanitization

Authentication Layer

- **Firebase Auth:** OAuth and email/password authentication
- **Token Verification:** Server-side JWT validation
- **Session Management:** Secure token handling

Database Layer

- **PostgreSQL:** ACID-compliant relational database
- **Row Level Security:** Per-row access control
- **Service Role:** Backend-only write access
- **Anonymous Role:** Frontend read access (select only)

Storage Layer

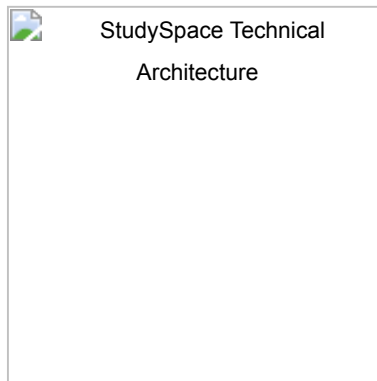
- **Cloudinary:** Optimized image delivery
- **Transformation:** On-the-fly image resizing
- **CDN:** Global edge caching

Security Layer

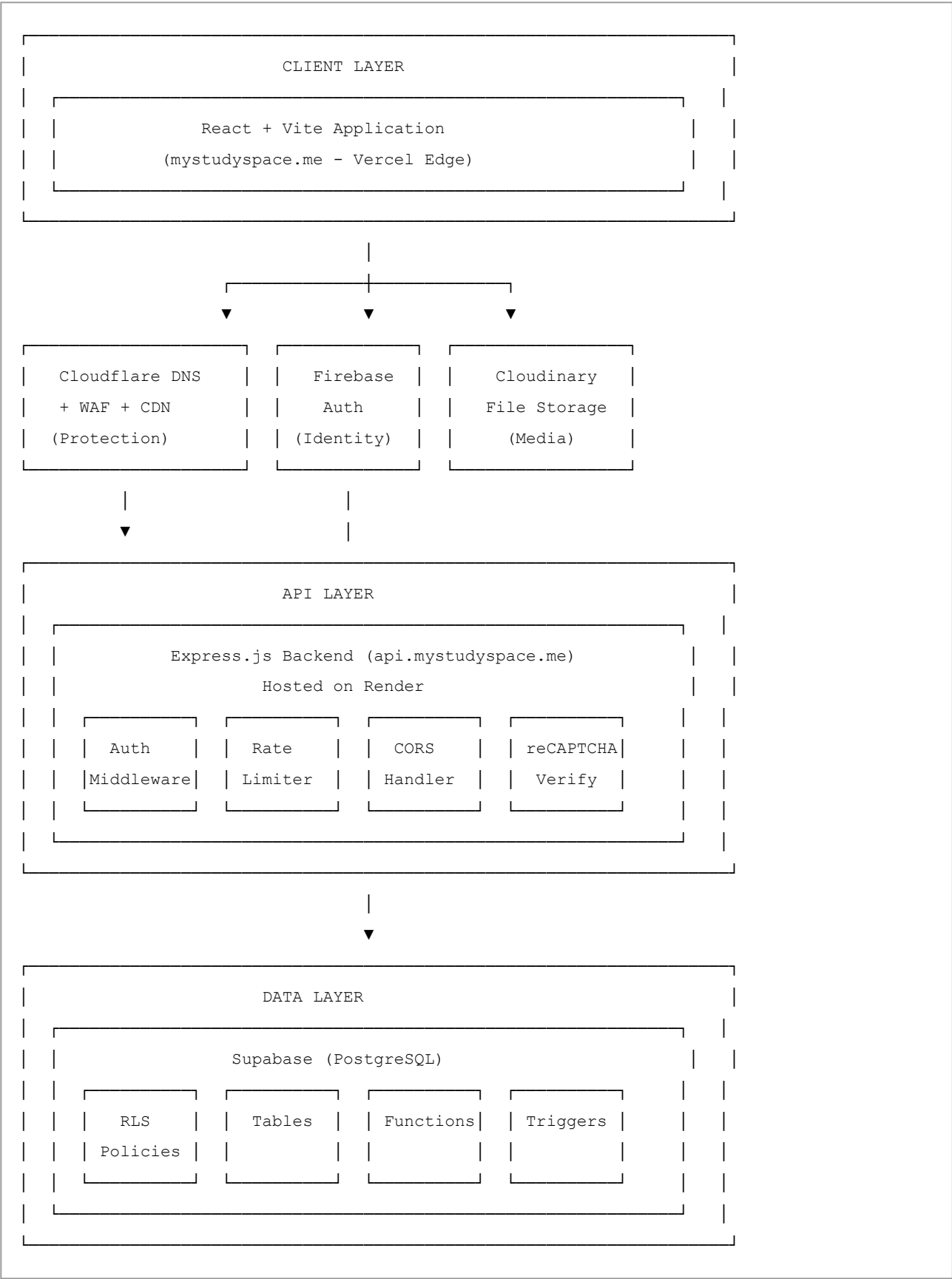
- **Cloudflare:** DDoS protection, WAF
 - **reCAPTCHA v3:** Bot prevention
 - **HTTPS:** End-to-end encryption
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6. Architecture Diagram

Visual Architecture



Textual Architecture Diagram



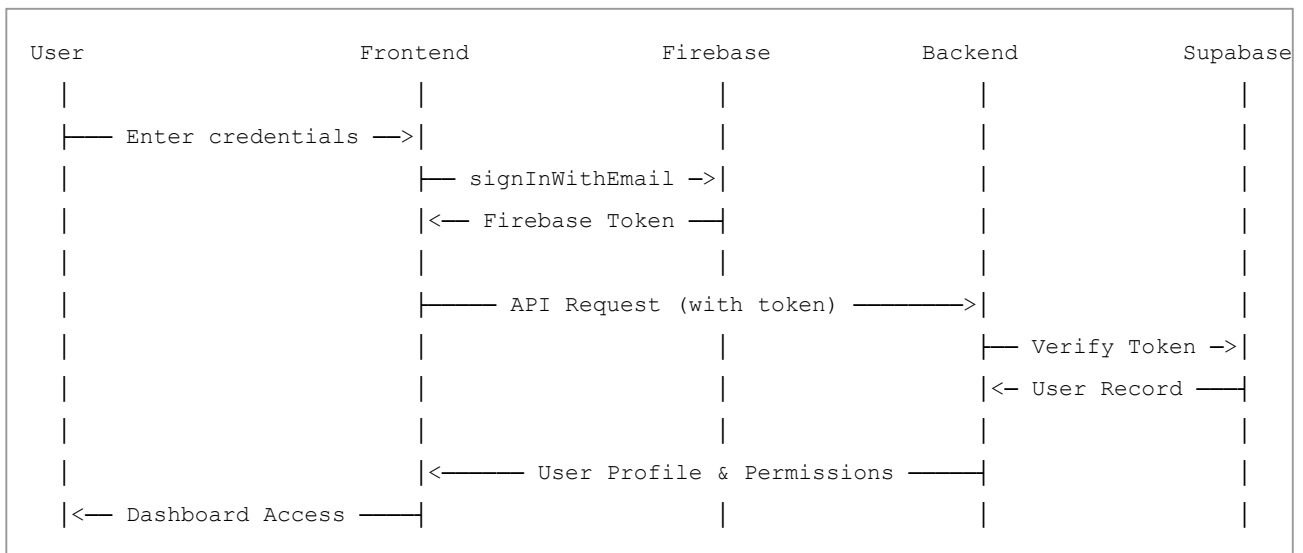
Component Interaction Summary

- 1. User Request Flow:** Browser → Cloudflare → Vercel CDN → React App
- 2. API Request Flow:** React App → Cloudflare → Render → Express API

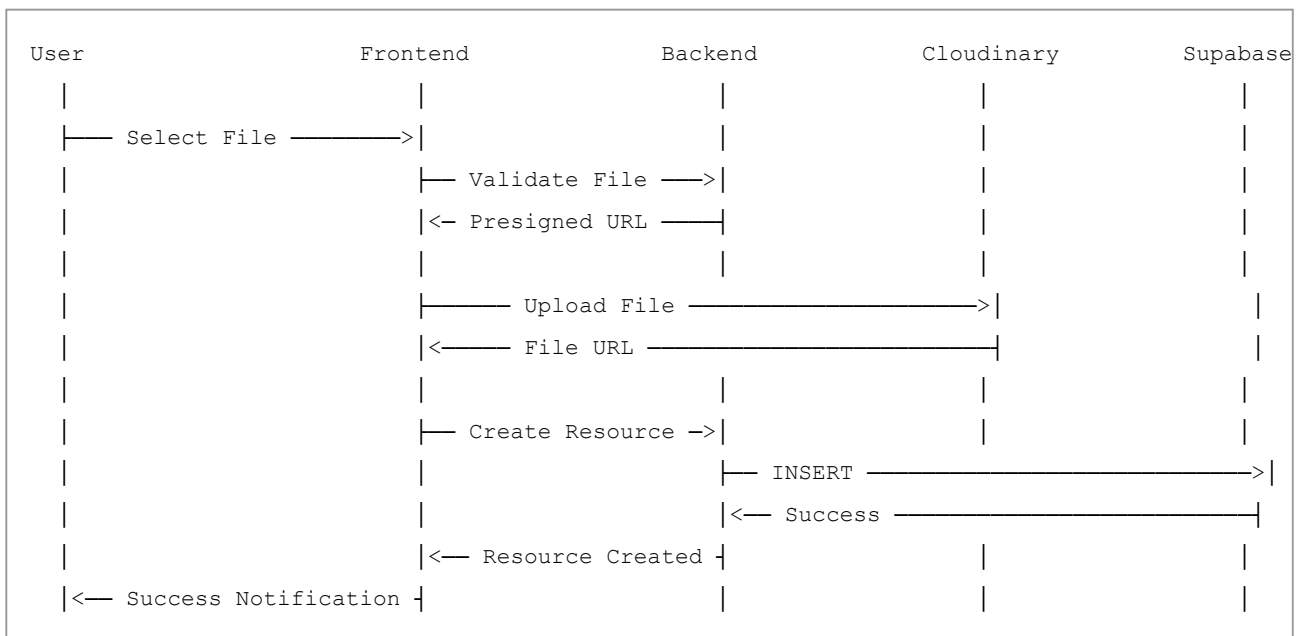
3. **Database Flow:** Express API (Service Role) → Supabase PostgreSQL
4. **Auth Flow:** React App → Firebase Auth → Express API (Token Verification)
5. **File Flow:** React App → Express API → Cloudinary

7. Data Flow

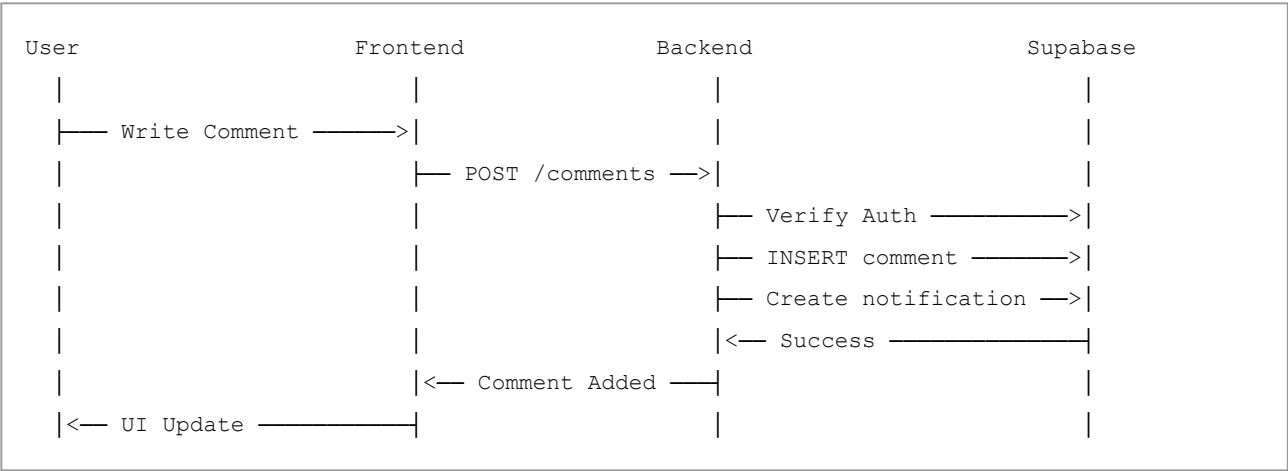
7.1 Authentication Flow



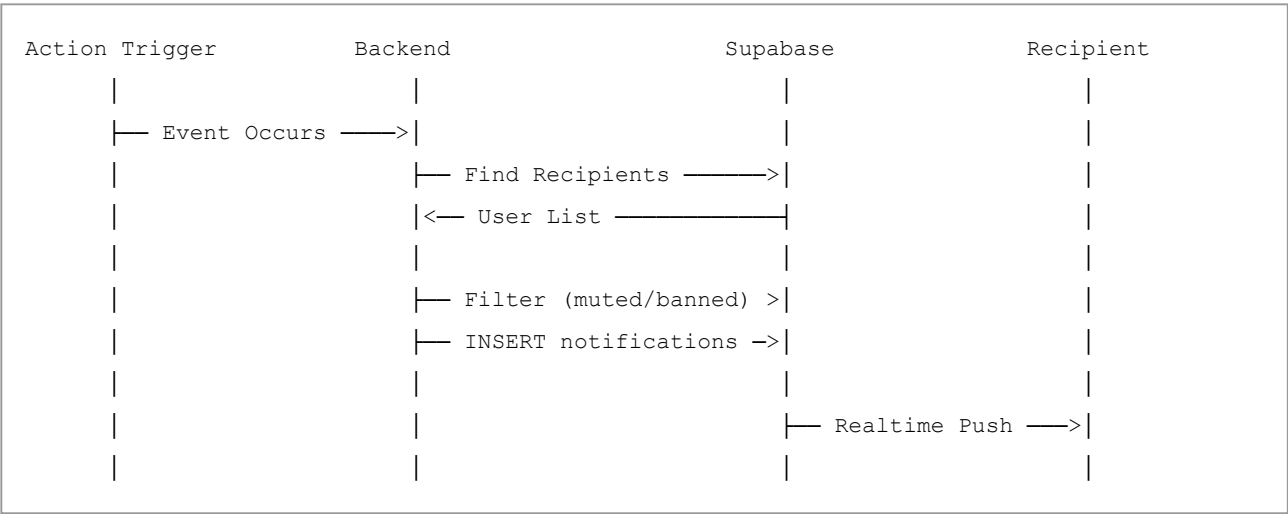
7.2 Resource Upload Flow



7.3 Comment & Reply Flow



7.4 Notification Flow



8. Database Design Overview

Core Tables

users

Primary user identity and profile storage.

- **id:** UUID (Firebase UID)
- **email:** Unique email address
- **display_name:** User's display name
- **username:** Unique username for profile URLs
- **profile_photo_url:** Base64 or URL of profile image
- **college:** Associated institution
- **bio:** User biography
- **created_at:** Account creation timestamp

resources

Academic content uploads.

- **id**: UUID
- **title**: Resource title
- **type**: notes | video | pyq
- **file_url**: Cloudinary URL
- **semester, branch, subject, chapter, topic**: Academic categorization
- **upvotes, downvotes**: Community voting
- **uploaded_by_email**: Creator reference
- **college_id**: Institution scope
- **is_approved**: Moderation status

chat_rooms

Discussion room containers.

- **id**: UUID
- **name**: Room name
- **description**: Room purpose
- **is_private**: Access control flag
- **join_code**: 6-character unique code
- **created_by_email**: Creator reference
- **member_count**: Cached member count

room_members

User-room associations.

- **room_id, user_email**: Composite key
- **user_name**: Display name cache
- **is_banned**: Moderation flag
- **notifications_muted**: User preference
- **joined_at**: Membership timestamp

chat_messages

Room message content.

- **id**: UUID
- **room_id**: Parent room
- **author_email, author_name**: Sender info
- **content**: Message text
- **image_url**: Optional attachment
- **upvotes, downvotes**: Message voting
- **parent_id**: Threading support

notifications

User notification queue.

- **id:** UUID
- **user_email:** Recipient
- **type:** follow_request | comment_reply | new_post | etc.
- **title, message:** Display content
- **is_read:** Read status
- **created_at:** Timestamp

bookmarks

User saved content.

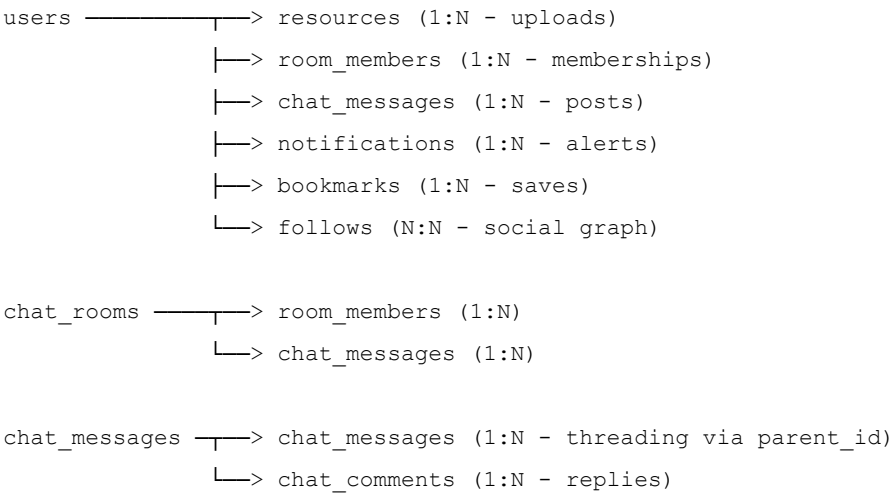
- **user_email, resource_id:** Composite key
- **created_at:** Save timestamp

follows

Social graph for users.

- **follower_email, following_email:** Relationship pair
- **status:** pending | accepted | rejected

Relationship Diagram (Conceptual)



9. Security Architecture

9.1 Authentication Strategy

Component	Implementation
Identity Provider	Firebase Authentication
Supported Methods	Email/Password, Google OAuth
Token Format	Firebase ID Token (JWT)
Token Verification	Server-side via Firebase Admin SDK

Component	Implementation
Session Duration	1 hour (auto-refresh)

9.2 Authorization Model

Frontend Authorization:

- Permission hooks (`usePermissions`) check user role
- UI elements conditionally rendered based on access
- Route guards redirect unauthorized users

Backend Authorization:

- Every API endpoint verifies Firebase token
- User role extracted from token claims
- Permission checks before database operations

Database Authorization:

- Row Level Security (RLS) on all sensitive tables
- Service role key used only by backend
- Anonymous key allows SELECT on public tables only

9.3 Row Level Security Policies

```
-- Example: Users can only update their own profile
CREATE POLICY "users_update_own" ON users
  FOR UPDATE USING (auth.uid()::text = id);

-- Example: Resources visible to same college
CREATE POLICY "resources_select" ON resources
  FOR SELECT USING (
    college_id = current_setting('app.college_id', true)
    OR is_approved = true
  );
```

9.4 Rate Limiting

Endpoint Type	Limit	Window
Authentication	10 requests	15 minutes
Resource Upload	20 requests	1 hour
Comments	30 requests	1 minute
General API	100 requests	1 minute

Implementation uses sliding window algorithm with Redis-compatible in-memory store.

9.5 reCAPTCHA Integration

- **Version:** reCAPTCHA v3 (invisible)
- **Protected Actions:** Login, signup, resource upload
- **Score Threshold:** 0.5 (configurable)
- **Fallback:** Challenge on low scores

9.6 File Upload Security

Check	Implementation
File Type	MIME type validation
File Size	10MB limit for documents, 50MB for videos
Content Scan	Planned: VirusTotal integration
Access Control	Signed URLs (in progress)

9.7 API Protection

- **CORS:** Whitelist of allowed origins only
- **HTTPS:** Enforced on all endpoints
- **Input Sanitization:** XSS prevention on all inputs
- **SQL Injection:** Parameterized queries via Supabase client
- **Request Validation:** Joi/Zod schema validation

9.8 Threat Prevention Matrix

Threat	Mitigation
DDoS	Cloudflare protection + rate limiting
Credential Stuffing	Rate limiting + reCAPTCHA
XSS	Content Security Policy + input sanitization
CSRF	Same-origin policy + token validation
Data Exfiltration	RLS + minimal data exposure
Privilege Escalation	Server-side role verification

10. Performance Architecture

10.1 Frontend Optimization

Code Splitting:

- Route-based lazy loading with React.lazy()
- Dynamic imports for heavy components (PDFViewer, ImageCropper)
- Vendor chunk separation

Bundle Optimization:

- Tree shaking enabled
- Minification and compression
- Source map generation for debugging only

Rendering Optimization:

- React.memo for expensive components
- useCallback/useMemo for stable references
- Virtualization for long lists (planned)

10.2 Data Fetching

React Query Configuration:

- Stale time: 5 minutes for resources
- Cache time: 30 minutes
- Background refetch on focus
- Deduplication of parallel requests

Optimistic Updates:

- Immediate UI updates for votes
- Rollback on failure

10.3 Caching Strategy

Layer	Technology	TTL
Browser	Service Worker (planned)	Variable
CDN	Cloudflare Edge	1 hour
API	In-memory cache	5 minutes
Database	Supabase connection pooling	Persistent

10.4 Asset Optimization

Asset Type	Optimization
Images	WebP conversion, lazy loading, srcset
Fonts	Preconnect, display swap
CSS	PurgeCSS, critical CSS inline
JavaScript	Minification, gzip compression

10.5 Mobile Performance

- Viewport-aware height units (dvh)
- Touch-optimized interactions
- Reduced animation on low-power mode
- Responsive images

11. User Flow

11.1 New User Journey

1. Landing Page

- User visits mystudyspace.me
- Views college selection screen

2. College Selection

- User searches for their college
- Selects from available institutions

3. Authentication

- User chooses signup method (email/Google)
- Enters credentials
- Receives verification email (if email signup)

4. Profile Setup

- User lands on Study page
- Can customize profile (name, photo, bio)
- Username auto-generated, can be changed

5. Onboarding

- Guided tour of features (planned)
- Suggested resources based on branch

11.2 Resource Discovery Flow

1. User navigates to Study page
2. Views resource grid with thumbnails
3. Filters by semester/branch/subject
4. Searches by keyword
5. Clicks resource card
6. Views resource details (votes, uploader, date)
7. Downloads or bookmarks resource

11.3 Resource Upload Flow

1. User clicks "Add Resource" button
2. Selects resource type (notes/video/PYQ)
3. Fills metadata form (semester, subject, etc.)
4. Uploads file via drag-drop or file picker
5. Submits for publishing
6. Resource appears in library (auto-approved or pending)

11.4 Commenting Flow

1. User opens resource or notice
2. Scrolls to comment section

3. Types comment in input field
4. Submits comment
5. Comment appears in thread
6. Other users can reply (nested threading)

11.5 Chat Room Flow

1. User navigates to Chat Rooms
 2. Views list of joined rooms
 3. Creates new room or joins via code
 4. Enters room
 5. Posts messages, images
 6. Votes on messages
 7. Can save/bookmark messages
-

12. Privacy & Data Protection

Data Collection Statement

StudySpace collects the following categories of personal information:

Account Information:

- Email address (required for authentication)
- Display name (user-provided)
- Profile photo (optional)
- College affiliation (required for access)

Activity Data:

- Resources uploaded
- Comments and messages posted
- Voting activity
- Bookmark history

Technical Data:

- IP address (for security and rate limiting)
- Device type and browser (for compatibility)
- Timestamp of actions

Data Usage

Collected data is used exclusively for:

1. **Service Provision:** Enabling core platform functionality
2. **Security:** Preventing abuse, fraud, and unauthorized access

3. **Improvement:** Analyzing usage patterns to enhance features
4. **Communication:** Sending essential notifications and updates

Data Protection Measures

1. **Encryption:** All data transmitted via HTTPS/TLS 1.3
2. **Access Control:** Role-based permissions at application and database levels
3. **Anonymization:** Analytics use aggregated, non-identifiable data
4. **Retention:** Active data retained while account exists; deleted upon request
5. **Backups:** Encrypted daily backups with 30-day retention

Data Sharing Policy

StudySpace does not:

- Sell personal data to third parties
- Share data with advertisers
- Use data for purposes unrelated to the platform

StudySpace may share data:

- With service providers (Firebase, Supabase, Cloudinary) for operational purposes
- When required by law or legal process
- To protect platform security and user safety

User Rights

Users may request:

- Access to their personal data
- Correction of inaccurate information
- Deletion of their account and associated data
- Export of their data in machine-readable format

13. Platform Policies

13.1 Acceptable Use Policy

Users of StudySpace agree to:

Permitted Activities:

- Share original or legitimately obtained academic resources
- Engage in respectful academic discussions
- Provide constructive feedback through voting
- Report inappropriate content or behavior

Prohibited Activities:

- Upload copyrighted material without permission
- Share content that violates institutional policies
- Engage in harassment, bullying, or discrimination
- Attempt to bypass security measures
- Create multiple accounts to circumvent restrictions
- Use automated tools without authorization

13.2 Content Policy

Acceptable Content:

- Academic notes, summaries, and study guides
- Previous year question papers (where permitted)
- Educational videos and tutorials
- Relevant links to external educational resources

Prohibited Content:

- Examination answers or leaked papers
- Plagiarized or copied materials without attribution
- Spam, advertisements, or promotional content
- Offensive, violent, or illegal content
- Personal information of others without consent

13.3 Abuse Prevention

StudySpace employs multiple layers of abuse prevention:

1. Automated Detection:

- Rate limiting on all actions
- reCAPTCHA for critical operations
- Pattern detection for spam

2. Community Moderation:

- Voting system surfaces quality content
- Report functionality for users
- Room moderators for chat spaces

3. Administrative Oversight:

- Review queue for flagged content
- User ban/suspension capabilities
- Audit logs for accountability

13.4 Moderation Philosophy

StudySpace follows a graduated enforcement approach:

1. **Warning:** First-time or minor violations
2. **Temporary Restriction:** Repeated or moderate violations
3. **Permanent Ban:** Severe or persistent violations

Appeals can be submitted via the platform support system.

14. Scalability Considerations

User Scaling

Current Capacity:

- Single-college deployment (KIET)
- Hundreds of concurrent users supported

Scaling Strategy:

- Multi-tenant architecture ready
- College-based data partitioning
- Independent database connections per institution

File Storage Scaling

Current Approach:

- Cloudinary free tier with upgrade path
- On-the-fly transformations

Scaling Strategy:

- Migrate to signed uploads for security
- S3-compatible storage for large files
- CDN caching for popular resources

Traffic Scaling

Current Infrastructure:

- Vercel Edge Functions for frontend
- Render with auto-scaling for backend
- Supabase connection pooling

Scaling Strategy:

- Horizontal pod scaling on Render
- Read replicas for Supabase (if needed)

- Regional edge deployment

Cost Efficiency

Service	Free Tier	Paid Scaling
Vercel	100GB bandwidth	Pay per GB
Render	750 hours/month	Pay per hour
Supabase	500MB database	\$25/month for 8GB
Cloudinary	25GB bandwidth	Pay per GB
Firebase Auth	50K monthly users	\$0.01/user after

Database Optimization

- Indexed queries on frequently accessed columns
- Materialized views for complex aggregations (planned)
- Connection pooling via Supabase
- Query optimization with EXPLAIN analysis

15. Future Roadmap

Q1 2026 (Current Quarter)

Feature	Status	Priority
Profile photo cropper	☐ Complete	High
Room settings (ban, mute, delete)	☐ Complete	High
Mobile UI optimization	☐ Complete	High
Dynamic student count	☐ Complete	Medium

Q2 2026

Feature	Description	Priority
Progressive Web App	Installable app with offline access	High
Push Notifications	Native browser notifications	High
Content Moderation Dashboard	Admin interface for content review	Medium
Analytics Dashboard	User engagement metrics	Medium

Q3 2026

Feature	Description	Priority
AI Resource Recommendations	ML-based content suggestions	High
Study Groups	Scheduled study sessions	Medium
Calendar Integration	Academic calendar sync	Medium
Multi-language Support	Hindi, regional languages	Low

Q4 2026

Feature	Description	Priority
Video Conferencing	Built-in study calls	High
Gamification	Badges, leaderboards	Medium
Institution Analytics	Reports for colleges	Medium
API for Third Parties	Developer access	Low

Long-term Vision

- Platform Expansion:** 100+ colleges across India
- Mobile Apps:** Native iOS and Android applications
- AI Tutoring:** Interactive doubt resolution
- Publisher Partnerships:** Official content integrations
- Certification System:** Skill verification badges

Appendix

A. Technology Dependencies

Category	Package	Version
Frontend Framework	React	18.x
Build Tool	Vite	5.x
UI Components	shadcn/ui	Latest
Styling	Tailwind CSS	3.x
State Management	React Query	5.x
Routing	React Router	6.x
Backend Framework	Express	4.x
Database Client	@supabase/supabase-js	2.x
Authentication	Firebase Admin	11.x
File Storage	Cloudinary	1.x

B. Environment Variables

Variable	Purpose	Required
VITE_SUPABASE_URL	Supabase project URL	Yes
VITE_SUPABASE_ANON_KEY	Public API key	Yes
VITE_FIREBASE_CONFIG	Firebase configuration	Yes
SUPABASE_SERVICE_ROLE_KEY	Backend-only key	Yes
CLOUDINARY_URL	File upload endpoint	Yes
RECAPTCHA_SECRET_KEY	Server-side verification	Yes

C. API Endpoints Summary

Method	Endpoint	Description
POST	/api/auth/verify	Verify Firebase token
GET	/api/resources	List resources

Method	Endpoint	Description
POST	/api/resources	Create resource
POST	/api/chat/rooms	Create chat room
GET	/api/chat/rooms/:id/messages	Get room messages
POST	/api/notifications	Create notification
GET	/api/follow/status	Check follow status

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